IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-SE-040

1. Topic (12 words max)

Emotionally Intelligent Companion Dog Robot with AI-Driven Security and Voice Interaction

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	NULL
Year	NULL

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Nature/Gravity of the Problem:

The project addresses the issue of loneliness, where many individuals suffer from a lack of companionship. The target user persona includes individuals seeking emotional comfort in a home environment. Furthermore, existing home security solutions often lack interactivity, acting merely as passive recording devices rather than active guardians.

Impact of the Problem:

- ***Social/Economic:*** *Advanced robotic companions that could solve this issue, such as the Sony Aibo, are prohibitively expensive and largely unavailable in markets like Sri Lanka.*
- ***Technical:*** *Current voice recognition systems rely heavily on cloud processing, leading to latency issues and privacy concerns regarding data transmission.*
- ***Security:*** *Standard security systems are prone to false positives and cannot distinguish between a friendly owner and a stranger in a meaningful, interactive way.*

Sustainable Development Goals:

The project aims to provide an affordable, locally built alternative that enhances well-being (SDG 3) and fosters innovation in affordable robotics (SDG 9).

6. Brief description of Existing Research and Systems (300 words max)

Summary of Existing Products/Systems:
Current market solutions include high-end robotic pets like the Sony Aibo and standard cloud-based voice assistants like Google Assistant. Traditional home security systems utilize motion sensors and cameras but lack autonomous mobility or social interaction.

Gaps/Limitations Identified:

- ***Cost & Availability:*** *Advanced robots are too expensive for the average user in the local context.*
- ***Latency & Privacy:*** *Voice recognition systems often depend on the internet, causing delays and potential privacy breaches.*
- ***False Positives in Security:*** *Mobile robots face challenges with uncontrolled lighting and motion blur, leading to recognition failures.*
- ***Lack of Emotional Intelligence:*** *Most security robots cannot detect human moods or adapt their behavior to provide comfort.*

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

Addressal of Gaps: *The solution is "Buddy," a wheeled robot dog that integrates emotional support with practical security. Unlike expensive imported models, Buddy is built using affordable components like Raspberry Pi and ESP32. It addresses privacy and latency by utilizing offline voice recognition and local processing for emotion and face detection.*

System Perspective (Input, Process, Output):

- ***Input:***
  - ***Audio:*** *Captured via a microphone array for voice commands.*
  - ***Visual:*** *Captured via an onboard camera for face and emotion recognition.*
  - ***Sensors:*** *Ultrasonic sensors capture distance data for obstacle avoidance.*
- ***Process:***
  - ***Voice Module:*** *Uses a Wake-Word engine (Porcupine) followed by offline Speech-to-Text (Vosk/Whisper) and NLU parsing to determine intent.*
  - ***Vision Module:*** *Uses FaceNet embeddings to calculate Euclidean distance for identity verification (Owner vs. Stranger). It employs a Mini-Xception CNN to analyze facial features for emotion detection.*
  - ***Navigation Module:*** *A Hybrid Control System combines Fuzzy Logic for safety (obstacle avoidance) and PID controllers for target tracking.*
- ***Output:***
  - ***Actuation:*** *Motor movements (navigation), head tilting, and tail wagging based on emotional state.*
  - ***Alerts:*** *If a stranger is detected, the system triggers a Security Guard Mode that sends a photo alert to the user's phone via Telegram.*
  - ***Feedback:*** *Audio responses or LED indicator changes.*

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

Acquiring Specialized Domain Expertise: *The team will leverage Deep Learning frameworks specifically optimized for edge devices. Expertise in Computer Vision will be applied through the use of Convolutional Neural Networks (CNN) and SSD models for object detection. The team will utilize Fuzzy Logic Control (FLC) and PID algorithms for the navigation subsystems.*

Technical Knowledge & Skills:

- ***Hardware:*** *Assembly of a physical wheeled robot using Raspberry Pi 4 and ESP32 microcontrollers.*
- ***Software:*** *Python programming for the Raspberry Pi, implementation of spectral subtraction for audio denoising, and integration of the Telegram API for alerts.*

Data Collection:

- ***Faces:*** *The "Labeled Faces in the Wild" (LFW) dataset will be used to benchmark the security module.*
- ***Emotions:*** *The FER2013 (Facial Emotion Recognition) dataset will be used to train the emotion detection CNN.*
- ***Validation:*** *A local dataset of team members' images and voices (Sri Lankan English accents) will be created to test real-world accuracy.*

9. Objectives and Novelty

Main Objective *To build an affordable, emotionally intelligent robot dog that provides companionship through emotion detection and voice interaction while ensuring home safety through autonomous security patrolling.*			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Ashandth Uthayashankar *IT22272522*	*Offline Voice Command Recognition*	1. *Implement Wake-Word detection using **Porcupine** or **Snowboy** algorithms to listen for the trigger "Buddy".* 2. *Deploy offline Speech-to-Text using **Vosk** or **OpenAI Whisper** base models.* 3. *Develop NLP parsing to map text to intents (e.g., cmd_navigation) using keyword extraction.*	*Implementation of a **completely offline** two-stage pipeline that ensures user privacy and eliminates cloud latency, optimized for local Sri Lankan English accents.*
Sobiya Anton Suresh IT22203694	*Face Recognition for Security & Familiarity*	1. *Implement **SSD (Single Shot Detector)** for rapid face detection and cropping.* 2. *Utilize **FaceNet** to generate 128-dimensional embeddings and calculate Euclidean distance for identity verification.* 3. *Integrate **API** to send photo alerts if a "Stranger" is detected in App.*	*A dual-purpose vision system that uses Deep Learning (FaceNet) on edge hardware to act as both a **biometric security guard** and a **companion recognizer***

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Member Name with Registration No	Sub Objective	Tasks	Novelty
SSY Wickramasinghe IT21816086	*Autonomous Navigation and Obstacle Avoidance*	*1. **Develop a Reactive Safety Layer** on the ESP32 using **Fuzzy Logic** for smooth, non-linear obstacle avoidance.* *2. **Implement Visual Servoing** on the Raspberry Pi using **OpenCV** and **PID Control** for stable face tracking.* *3. **Create a Subsumption Architecture** via UART to prioritize safety reflexes over high-level navigation commands.*	Utilization of a Hybrid Control System (Fuzzy Logic + PID) within a priority-based arbitration architecture, allowing the robot to survive in unstructured dynamic environments.
We don't have a 4th member as we 3 will be working on it	*Real-Time Facial Emotion Recognition*	1. *Implement the **Mini-Xception CNN** model optimized for resource-constrained edge devices.* 2. *Train the model using the **FER2013** dataset to detect "Happy," "Sad," and "Neutral" states.* 3. *Develop the logic to trigger behavioral states (e.g., Playful_Mode, Comfort_Mode) based on the detected emotion*	*Deployment of a lightweight **Convolutional Neural Network** capable of detecting subtle micro-expressions in real-time on a Raspberry Pi without requiring heavy cloud processing.*

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10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

Had discussed Proposal and adjust each components

	Title	First Name	Last Name	Signature
Supervisor	Prof	Pradeep	Abeygunawardhana	
Co-Supervisor	Prof	Samarth	Rajapaksha	
External Supervisor	Eng. (Senior Data Scientist / Machine Learning Engineer)	Rishikesan	Srikusan	

Summary of external supervisor's (if any) experience and expertise:
Over 9 years of experience in engineering and applied AI, with expertise in machine learning, computer vision, and data-driven system development.